The set Tot of indices of total computable functions is not computable

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-computability-computability-theory/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source's sentence 'the we compute~h' is the author's own typo and is delivered verbatim

Editorial note. The label prop:total names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- The halting set K. K = {x : phi_x(x) converges} is computably enumerable but not computable.. K reduces to Tot; non-computability of K transfers along the reduction.
- s-m-n theorem. For each e there is a primitive recursive k(x) with phi_{k(x)}(y) equal to a prescribed partial computable function of (x,y).. Freezes y out: k(x) indexes the function that is 0 when phi_x(x) halts and undefined otherwise.

Proposition 0.1. Tot *is not computable.*

Proof. To see that Tot is not computable, it suffices to show that K is reducible to it. Let $h(x, y)$ be defined by

$$h(x, y) \simeq \begin{cases} 0 & \text{if } x \in K \\ \uparrow & \text{otherwise} \end{cases}$$

Note that $h(x, y)$ does not depend on y at all. It should not be hard to see that h is partial computable: on input x, y, the we compute h by first simulating the function φ_x on input x; if this computation halts, $h(x, y)$ outputs 0 and halts. So $h(x, y)$ is just $0(\mu s T(x, x, s))$, where 0 is the constant zero function.

Using the *s-m-n* theorem, there is a primitive recursive function $k(x)$ such that for every x and y,

$$\varphi_{k(x)}(y) = \begin{cases} 0 & \text{if } x \in K \\ \uparrow & \text{otherwise} \end{cases}$$

So $\varphi_{k(x)}$ is total if $x \in K$, and undefined otherwise. Thus, k is a reduction of K to Tot. □